

GERA LEUMNITZ, Germany

WMO: 105670

Lat: 50.8814N

Lon: 12.1289E

Elev: 1015

StdP: 14.16

Time Zone: 1.00 (EUC)

Period: 82-03

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	6.0	11.5	2.1	6.3	8.5	7.3	8.3	12.4	31.1	41.3	26.9	40.1	5.0	40	0.599

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.4	83.8	66.0	80.2	64.6	77.0	63.2	67.8	79.5	65.9	76.4	64.4	74.4	7.9	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.7	91.8	72.1	62.1	86.6	70.0	60.7	82.3	68.4	32.7	79.8	31.2	76.2	30.0	74.5	75.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 24.8	21.6	18.9	DB	2.0	90.7	12.2	2.8	-6.7	92.7	-13.8	94.4	-20.7	95.9	-29.5	97.9	(4)
(5)			WB	1.5	71.6	11.8	2.3	-7.0	73.3	-13.9	74.6	-20.6	75.9	-29.2	77.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	47.2	31.5	30.6	38.7	44.9	53.8	59.2	63.7	63.7	57.0	48.8	39.0	34.5
	DBStd	13.87	9.98	10.17	7.90	6.85	6.56	6.31	5.77	6.04	5.93	7.18	6.98	7.37
	HDD50	2626	575	545	354	183	38	3	0	0	7	108	332	481
	HDD65	6630	1040	964	814	603	347	193	97	98	247	502	779	946
	CDD50	1614	1	1	5	29	157	280	423	425	218	71	3	1
	CDD65	143	0	0	0	0	1	20	55	58	8	1	0	0
	CDH74	1424	0	0	0	1	40	216	549	529	85	4	0	0
	CDH80	331	0	0	0	0	0	43	137	144	7	0	0	0
Wind	WSAvg	8.6	11.0	9.8	9.6	8.1	7.2	7.1	6.7	6.8	7.7	8.8	9.9	10.1
Precipitation	PrecAvg	24.6	1.4	1.3	1.6	1.5	2.4	2.8	3.3	2.6	2.2	1.7	2.0	1.7
	PrecMax	33.6	2.7	2.2	3.8	3.1	5.9	5.1	5.5	7.1	5.2	3.6	5.1	3.6
	PrecMin	17.2	0.0	0.2	0.2	0.2	0.4	0.8	0.7	0.6	0.5	0.7	0.0	0.6
	PrecStd	4.3	0.7	0.6	0.9	0.7	1.2	1.1	1.3	1.6	1.1	0.7	1.1	0.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	51.7	57.0	63.6	70.7	77.8	85.1	88.1	89.9	80.9	72.7	58.5	53.3
		MCWB	47.2	49.3	51.0	55.0	61.7	66.5	68.7	67.6	63.3	60.3	51.4	48.5
	2%	DB	47.3	51.0	57.1	65.5	74.8	79.8	84.0	83.7	76.4	66.6	53.9	50.3
		MCWB	44.1	45.7	49.0	53.3	59.9	64.7	65.9	66.2	62.3	57.0	49.4	46.6
	5%	DB	45.1	46.4	52.4	60.9	70.6	75.7	80.2	79.7	72.2	62.5	51.0	47.7
		MCWB	42.3	42.5	46.1	50.9	58.1	62.9	64.9	64.3	60.7	55.3	47.5	44.7
	10%	DB	42.7	41.9	49.0	56.8	67.0	71.6	76.5	76.1	68.1	58.9	48.4	44.7
		MCWB	39.9	38.9	44.3	49.0	56.3	60.4	63.4	63.0	59.2	53.4	45.2	42.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	47.4	50.0	52.7	56.5	64.2	68.7	70.5	71.0	66.0	61.8	52.4	49.3
		MCDB	50.7	55.1	61.0	67.7	74.6	80.4	84.6	84.8	75.5	70.9	56.3	51.7
	2%	WB	44.7	45.9	49.4	54.0	61.1	65.7	68.1	67.5	63.6	58.4	50.0	47.0
		MCDB	47.1	50.0	56.3	62.9	71.2	76.5	79.5	79.8	73.2	64.8	53.4	50.0
	5%	WB	42.7	42.6	46.9	51.7	59.3	63.9	66.1	65.6	61.8	55.9	47.8	44.9
		MCDB	44.9	46.2	51.4	59.6	68.7	73.4	76.2	76.1	70.2	61.4	50.8	47.6
	10%	WB	40.2	39.3	44.7	49.3	57.2	61.9	64.3	64.0	59.9	53.6	45.5	42.5
		MCDB	42.5	41.8	48.8	55.7	65.2	70.0	74.2	73.6	66.8	57.7	48.2	44.8
Mean Daily Temperature Range	5% DB	MDBR	9.0	11.1	12.5	15.2	17.7	16.5	18.4	18.1	16.0	14.1	10.4	7.8
		MCDBR	10.1	15.0	18.7	22.7	24.6	24.3	26.4	24.9	23.3	19.3	14.0	10.0
		MCWBR	8.3	10.5	12.2	12.6	12.7	11.4	11.5	10.6	11.5	11.4	10.3	8.2
	5% WB	MCDBR	9.7	14.7	17.3	20.8	21.9	21.9	23.0	21.4	20.3	17.6	13.0	9.6
		MCWBR	8.4	10.6	11.8	12.0	12.2	11.1	11.0	10.2	10.5	10.9	10.2	8.1
Clear-Sky Solar Irradiance	taub	0.305	0.328	0.370	0.405	0.408	0.406	0.422	0.413	0.379	0.355	0.328	0.300	
	taud	2.390	2.351	2.270	2.218	2.242	2.273	2.234	2.276	2.355	2.405	2.418	2.392	
	Ebn at Noon	229	253	261	264	269	270	263	259	256	240	217	212	
	Edh at Noon	20	26	35	41	42	41	42	38	31	25	19	17	
All-Sky Solar Radiation	RadAvg	245	479	799	1275	1538	1663	1576	1364	979	574	272	187	
	RadStd	40	74	107	164	184	141	187	131	133	70	36	22	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (54 neighbors)	+0.70	N/A	N/A	N/A	N/A	+0.59	+0.67	N/A	N/A	+41

Nomenclature: See separate page